

# RADIATION

28–31 Matter

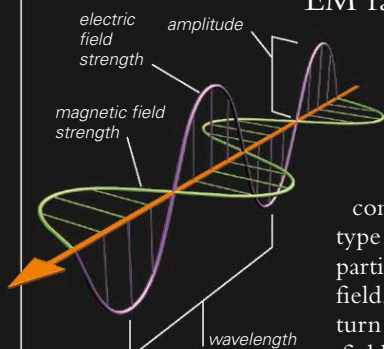
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**RADIATION IS ENERGY IN THE FORM** of waves or particles that are emitted from a source and can travel through space and some types of matter. Electromagnetic (EM) radiation includes light, X-rays, and infrared radiation. Particulate radiation consists of fast-moving charged particles such as cosmic rays and particles emitted in radioactive decay. EM radiation is vastly more significant in astronomy.



## HOW WAVES TRAVEL

An EM wave consists of oscillating electrical and magnetic fields arranged perpendicular to each other and carrying energy forward.

## ELECTROMAGNETIC RADIATION

Energy in the form of EM radiation is one of the two main components of the universe, the other being matter (see p.28). This type of radiation is produced by the motion of electrically charged particles, such as electrons. A moving charge gives rise to a magnetic field. If the motion is constant, then the magnetic field varies and in turn produces an electric field. By interacting with each other, the two fields sustain one another and move through space, transferring energy. As well as visible light, EM radiation includes radio waves, microwaves, infrared (heat) radiation, ultraviolet radiation, X-rays, and gamma rays. All these phenomena travel through space at the same speed—called the velocity of light. This speed is very nearly 186,000 miles (300,000 km) per second or 670 million mph (1 billion kph).

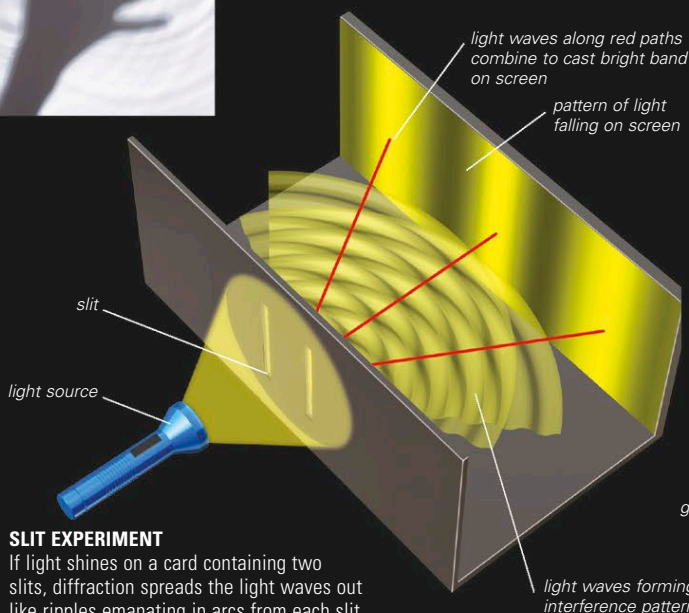
## WAVELIKE BEHAVIOR

In most situations, EM radiation acts as a wave—a disturbance moving energy from one place to another. It has properties such as wavelength (the distance between two successive peaks of the wave) and frequency (the number of waves passing a given point each second). This wavelike nature is shown by experiments such as the double-slit test (see below), in which light waves diffract (spread out) after passing through a slit and

also interfere with each other as their peaks and troughs overlap. The forms of EM radiation differ only in wavelength, but this affects other properties, such as penetrating power and ability to ionize atoms (see p.28).

### INTERFERING WAVES

The slit experiment is analogous to disturbing two points on the surface of a liquid. The ripples interfere to corrugate the liquid's surface.

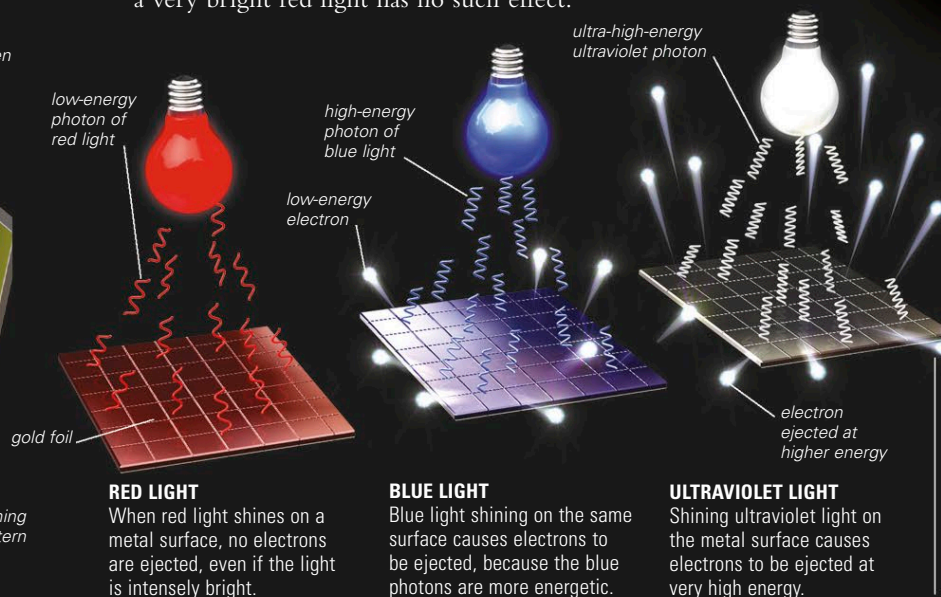


### SLIT EXPERIMENT

If light shines on a card containing two slits, diffraction spreads the light waves out like ripples emanating in arcs from each slit. The two wave trains then interfere to produce a light and dark pattern on the screen.

## PARTICLELIKE BEHAVIOR

EM radiation behaves mainly like a wave, but it can also be considered to consist of tiny packages or “quanta” of energy called photons. Photons have no mass but carry a fixed amount of energy. The energy in a photon depends on its wavelength—the shorter the wavelength, the more energetic the photon. For example, photons of blue (short-wavelength) light are more energetic than photons of red (long-wavelength) light. A classic demonstration of light’s particlelike properties is provided by a phenomenon called the photoelectric effect (see below). If a blue light shines on a metal surface, it causes electrons to be ejected from the metal, whereas even a very bright red light has no such effect.



### RED LIGHT

When red light shines on a metal surface, no electrons are ejected, even if the light is intensely bright.

### BLUE LIGHT

Blue light shining on the same surface causes electrons to be ejected, because the blue photons are more energetic.

### ULTRAVIOLET LIGHT

Shining ultraviolet light on the metal surface causes electrons to be ejected at very high energy.